UNIVERSITY OF GHANA, LEGON



Focused Attention and Recall of Auditory Information in Single- and Dual-Task Conditions

INDEX NUMBER

11360015

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ABSTRACT

There is evidence that controlled attention has some effect on the deep processing of single and dual auditory tasks, and the purpose of the present study was to investigate the effect of controlled attention on single and dual auditory materials. Subjects (N = 320) were selected from University of Ghana students and grouped into four groups in a between-subjects design. Subjects in CG1 and CG2 attended to only one of the two passages read by the researcher or the experimenter, and also answered ten questions from either passage 1 or passage II. Subjects in the experimental groups (EG1 and EG2) attended to two passages read by themselves and the experimenter simultaneously and were asked to answer 10 questions from the passage, either read by themselves or by the experimenter. The total number of correct responses was analysed using the ANOVA and multiple analyses. The results showed that CG1 subjects performed better than both EG1 and EG2 subjects. Moreover, CG2 performed better than EG1 and EG2.. EG1 and CG1 also performed better than EG2 and CG2, respectively. Findings imply that there is a need to control attention for both single and dual auditory information to increase memory retention.

Introduction and Literature Review

In the very earliest years of psychology, cognitive psychologists have been interested in how we notice certain things rather than others, and what is happening when we pay attention to something. Best (1995) defined attention as the concentration and focusing of mental effort - a focus that is selective, shiftable and divisible. By this, we can say that it is possible for us to select what to focus our attention on, move our attention from one object to another and also concentrate attention on more than one object.

The human system is exposed to many stimuli from its environment, and so we constantly perform two tasks at the same time. It is therefore important to consider the effect that performing two tasks at the same time has on auditory material. To explain this process, Broadbent (1958) propounded the filter theory of attention, which states that the non-shadowed information is completely blocked from entering the short-term memory by an attentional filter and hence cannot be processed for meaning; the attended information or the information that does not exceed capacity, however, is allowed to go through the attentional filter. Thus, we have no information about the excess information or the information that exceeds capacity.

According to Cherry (1953), based on his experience, people are capable of shadowing a spoken message with few errors, but then they are also unable to report accurately about the unattended message. This means that shadowing is a means by which we are able to control our attention to the relevant materials. Treisman (1960), in her attenuation theory of attention, states that the unattended information is not completely blocked but that the volume is turned down (unattended). Thus, we can still recall some information from the unattended message. In another experiment by Moray

(1959) subjects failed to recognize in a pose the occurrence of most of the words in the unattended message even though the list had been repeated 35 (thirty-five) times.

Sinnett (2013) used a divided attention psychophysical task to test the hypothesis that auditory attention to a stimulus feature facilitates the processing of other stimuli sharing the same feature. This attentional effect was explained based on the fact that the within-modality is dependent upon the presence of competing stimuli. The findings indicated that dual tasking led to lower performance compared to single tasking due to the presence of competing stimuli. Soto-Faraco (2015) found that single tasks lead to better performance compared to dual tasks. However, shadowing was found to increase the processing of the primary task in the dual tasking equally to the processing of the single auditory task.

The study by Kanneworff (2016) suggested that dual-tasking confusion occurs at the time of encoding memories, not during storage or retrieval. This would predict that source errors would originate at encoding, also, so we might expect to see the most errors in the simultaneous condition (S), when both auditory stimuli are available together. Troyer (2012), however, theorised that the degree to which an unrelated task would interfere with the source monitoring task would depend on the nature of the secondary task. Specifically, if the secondary task requires significant resources and controlled attention, it would interfere more than the simultaneous presentation of stimuli in the same modality, since the stimuli and their presentation modality are tightly bound.

Fairbank, Kingstone, Hillstrom and Shapiro (2015) found that a sample of 38 males and 34 females performed significantly better in single tasks than in dual tasks. They also found that shadowing increases performance for both single and dual tasks. Spence (2013), however, found that single tasking leads to deep processing and dual tasking is associated with shadow processing.

Aims of Study

The study conducted was to investigate the nature of the selection process in attention.
It was also aimed at assessing the fate, which is what happens to the unattended stimuli.

Statement of Hypotheses

CG1 will perform better than both EG1 and EG2
CG2 will perform better than both EG1 and EG2
Participants in control group 1 will perform better than participants in control group 2
The experimental group 1 participant will perform better than the experimental group 2 participants.

Operational Definition

Auditory stimulus: passages that were read by the subjects or the experimenter

Shadowed: To read passages aloud

Unattended auditory stimulus: a passage that participants did not perceive

Focused attention: Mentally focusing on one passage and leaving the other

METHODOLOGY

Population and sample

A sample of 320 undergraduate students was randomly selected from a homogeneous population of University of Ghana students offering PSYC 441/449 at the Accra city campus and the Legon campus. This was done using the simple random sampling technique. The selected participants were then assigned to four equivalent groups using the random assignment technique.

Instrument/materials

Materials used for the study included: two simple passages, 130 words each, containing 10 simple questions. Pens and scoring sheets were also used for recording the number of correct responses of students.

Design and procedure

The study used the completely independent group design. Subjects were randomly assigned to one of the four groups. Each group had an equal number of subjects. Those of CG 1 read passage 1 aloud and answer ten questions on passage 1. CG2 answered ten questions from passage 2, which was read by the experimenter aloud to them. EG 1 was asked to shadow passage 1 while the experimenter read out passage 2 simultaneously. Subjects answered ten questions from the shadowed passage 1. EG 1 read out passage 1 while the experimenter read out passage 2 simultaneously. The subjects in EG2 answered ten questions from passage 2. The number of questions correctly answered by all the subjects of all 4 groups was recorded as their respective scores

Scoring

Each correct response given attracted 1 mark, and a wrong response 0 marks.

RESULTS

TABLE 1: MEANS AND STANDARD DEVIATION OF THE SCORES AMONG THE FOUR GROUPS

Conditions	N	Mean	Std. Deviation	Variance
CG1	80	5.39	2.01	4.04
CG2	80	4.85	1.55	2.41
EG1	80	3.20	1.82	3.31
EG2	80	1.71	1.22	1.50
Total	320	3.79	2.21	4.88

EG1 mean correct response is 3.20, EG2 is 1.71, CG1 is 5.39, and CG2 is 4.85. The analysis of variance is used to find out if the group differences are significant.

TABLE 2: ONE-WAY ANOVA OF THE VARIOUS GROUPS

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	667.175	3	222.392	79.106	.000
Within Groups	888.375	316	2.811		
Total	1555.550	319			

The mean difference is significant ($F_{(3, 316)} = 79.106$, $p<.01$). The multiple comparisons were used to find the group that differs.

Table 3: Multiple Comparison of the Mean Scores among the four mean scores

Groups	CG1	CG2	EG1	EG2
CG1	-	0.53*	2.18*	3.68*
CG2	-	-	1.65*	3.13*
EG1	-	-	-	1.48*
EG2	-	-	-	-

*. The mean difference is significant at the 0.05 level

Results from Table 3 show the multiple comparisons of the means that differ significantly. From the table, comparing CG1 and CG2 to EG1 and EG2, CG2 mean score differed from the mean score of EG1 and EG2 ($p<.05$). The first hypothesis that states that CG2 will perform better than both EG1 and EG2 is supported.

CG2, however, differed in the performance of EG1 and EG2. The second hypothesis, which states that CG2 will perform better than both EG1 and EG2, is supported.

The difference between the two control groups is significant ($p<.05$). The third hypothesis that participants in control group 1 will perform better than participants in control group 2 is supported.

The fourth hypothesis that experimental group 1 participants will perform better than experimental group 2 participants was also supported.

DISCUSSION AND CONCLUSION

The study was set forth to investigate controlled attention in auditory processing. Four hypotheses were tested. The first hypothesis, which states that CG2 participants will perform better than both EG1 and EG2 participants, is supported. CG2 had a higher and significant mean score compared to subjects in EG1 and EG2. The finding indicates the importance of controlled attention on attended auditory materials. This also indicates that the unattended information is not able to be processed for full meaning, thus some important information may get lost as explained by the study conducted in the dichotic listening task (Cherry, 1953). The finding is in line with the study by Fairbank, Kingstone, Hillstrom and Shapiro (2015), who found that controlled processing increases performance and helps to overcome other contradictory stimuli. They found shadowing to increase performance for both single and dual tasking, as in the case of EG1.

The significant difference in performance between subjects in EG1 and CG2 can be explained by the dichotic listening task (Cherry, 1953). The dichotic listening task found that people are capable of shadowing a spoken message with few errors. This is what happened among the subjects in EG1. They committed more errors than those in CG2 because, even though attention was devoted to the passage, they attended to more tasks than they could process.

According to the dichotic listening task, when asked about the unattended spoken word, subjects were able to report some but not all. This is what happened to the EG2. Subjects could not attend to some of the information due to the simultaneous reading by the researcher and the subject, thus creating less correct responses because they were assessed on the unattended ear.

The second hypothesis, which states that CG1 will perform better than both EG1 and EG2, was supported. This means performing one task increases performance compared to performing two tasks at the same time if maximum attention is devoted to the single task. This indicates the importance of attending to one task at a time in auditory attention. This also indicates that the dual tasks could not process all the tasks for full meaning, thus some important information may get lost, as explained by the study conducted in the dichotic listening task (Cherry, 1953).

The high performance of the single task group (CG1) compared to the dual task conditions (EG1 and EG2) is in line with the study by Soto-Faraco (2015), which found that single tasks lead to better performance compared to dual tasks. The finding again is in line with the study by Fairbank, Kingstone, Hillstrom and Shapiro (2015) that found that a sample of 38 males and 34 females performed significantly better in single tasks than in dual tasks. The low performance of the dual

tasks groups can be due to the fact that subjects could not attend to some of the information due to the simultaneous engagement of the two tasks thus creating less correct response.

The study by Kanneworff (2016) suggested that dual-tasking confusion occurs at the time of encoding memories, not during storage or retrieval. This would predict that source errors would originate at encoding, also, so we might expect to see the most errors in the simultaneous condition (S), when both auditory stimuli are available together. Troyer (2012), however, theorised that the degree to which an unrelated task would interfere with the source monitoring task would depend on the nature of the secondary task, as occurred among subjects in EG1 and EG2.

The third hypothesis that participants in control group 1 will perform better than participants in control group 2 was supported. From the results, subjects in CG1 and CG2 differed significantly in performance, with participants in CG1 performing significantly higher than participants in CG2. The significant difference between subjects in CG1 and CG2 can be attributed to the type of processing adopted by the two groups. Participants in CG1 shadowed the information, which is a way of deeply processing information, whilst those in CG2 did not. This means participants in CG2 were unable to process the information as deeply as participants in CG1.

This finding is in line with the study Spence's (2013) study, which found that tasks deeply processed are better remembered that task shallowly processed. Everyone has a limited amount of resources, and at every point in time, there is a distraction which prevents people from processing information. Therefore, with controlled processing, the distractive information can be controlled, and that was what participants in CG1 did.

The fourth hypothesis that experimental group 1 participants will perform better than experimental group 2 participants was also supported. From the study, though subjects in EG1 and EG2 all engaged in dual tasks, which have been found to impede performance, participants in EG1 were assessed on the attended information, whilst those in EG2 were assessed on the unattended information. This finding is in agreement with the study by Fairbank, Kingstone, Hillstrom and Shapiro (2015), which found that shadowing increases performance for a dual task. The finding, however, agrees with the study by Spence (2013) that found that dual tasking is associated with shadow processing.

The findings are again in agreement with the study conducted by Kanneworff (2016), which suggested that dual tasking confusion occurs at the time of encoding memories, not during storage or retrieval. This would predict that source errors would originate at encoding also, so we might expect

to see the most errors in the simultaneous condition (S), when both auditory stimuli are available together, unless maximum attention is devoted to the information, as occurred among subjects in EG1.

Findings from this study have shown that we are capable of focusing attention on one or few tasks but usually the one we devote all our attentional resources on is remembered better. This can be applied to our classrooms and in our day-to-day activities, such as driving whilst talking, bathing whilst singing, etc. Drivers must cultivate the habit of devoting maximum cognitive resources to driving and avoid talking on the phone while driving. Students must desist from learning and listening to music at the same time. This can lead to cognitive overload and distraction.

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APPENDICES

Passage I

This is Morning News Magazine, and I am Jack Stevens. I'll be your host while Mark Osei is on assignment in the Western Region. Today's story is about flight from the cities. Everyone knows that it is happening, but only recently have we been able to determine where people are going. To the suburbs? To the fringes of the city? Surprisingly not.

In a marked reversal of Ghana's migration patterns, non-metropolitan areas have started growing faster than the metropolitan areas. City dwellers are leaving to settle in small towns in Ghana.

Surveys consistently show that a majority of people, including 4 out of 10 big-city dwellers, prefer life outside the urban environment. They associate small towns with a feeling of security. Tomorrow's report will focus on crime control. Till then, this is Jack Stevens wishing you a good morning.

Passage 1 Questions

1) Who was the program presenter on the Morning News Magazine? (Jack Stevens)
2) Where has the regular presenter gone? (On assignment in Western Region)
3) What is the topic of this talk? (Migration out of cities)
4) Where are many people moving? (To small towns)
5) Are people moving to suburbs? (No)
6) Do all city-dwellers prefer life in the urban environment? (No, 4 out of 10)
7) Do people feel safe in small towns? (Yes)
8) Are metropolitan areas growing faster than non-metropolitan areas? (NO)
9) At what time of day is the program presented? (Morning, a.m.)
10) What will tomorrow's report focus on? (Crime control)

Appendix 2: Passage II

Good morning. This is Danny Jackson with Weather Watch, brought to you at this time every Monday by the Sunyani Chamber of Commerce. The week-long extended forecast for Sunyani and Brong Ahafo Region calls for mostly sunny weather today and Tuesday, with temperatures in the high seventies. By Wednesday, a cold pressure area that has been building up over the Gulf of Guinea should begin to move inland over Brong Ahafo, bringing about a 10-degree drop in temperatures over the South Central of the Region.

By the middle of the week, that same low pressure should probably be dropping rain over the Region. By Friday afternoon, though, if all goes well, and throughout the weekend, the weather will be clear and warm. Danny Jackson with Weather Watch. Have a good week, Sunyani.

Passage II Questions

1. Who was the presenter of the weather forecast? (Danny Jackson)
2. Where is this report being broadcast? (In Sunyani, Brong Ahafo)
3. Is the duration of the forecast 24 hours? (No)
4. Is the pressure that has been building on the Gulf of Guinea hot or cold? (Cold)
5. Will the temperature in the southern part of the region rise or drop by 10˚? (drop)
6. What was the weather like on the day of the forecast? (Warm and sunny)
7. When will it be likely to rain? (In the middle of the week)
8. On what day of the week was the weather report forecast? (Monday)
9. At what time of day is the program presented? (Morning, a.m.)
10. What will the weather be like for the weekend? (Clear and warm)

Control GRP 1 (CG1)			CONTROL GRP 2 (CG2)			EXP GRP 1 (EG1)			EXP GRP 2 (EG2)	
SUBJECT	TOTAL		SUBJECTS	TOTAL		SUBJECT	TOTAL		SUBJECT	TOTAL
1	6		1	6		1	1		1	3
2	4		2	5		2	5		2	1
3	7		3	4		3	3		3	3
4	6		4	6		4	2		4	2
5	3		5	8		5	4		5	3
6	4		6	7		6	5		6	2
7	5		7	6		7	3		7	3
8	4		8	7		8	2		8	1
9	6		9	5		9	1		9	1
10	5		10	6		10	2		10	3
11	10		11	5		11	4		11	2
12	6		12	3		12	6		12	0
13	3		13	7		13	3		13	3
14	9		14	6		14	5		14	2
15	3		15	6		15	6		15	3
16	4		16	4		16	6		16	2
17	10		17	4		17	1		17	2
18	3		18	3		18	2		18	5
19	2		19	4		19	2		19	3
20	5		20	2		20	4		20	1
21	4		21	5		21	6		21	3
22	9		22	6		22	2		22	4
23	10		23	4		23	3		23	2
24	7		24	5		24	5		24	1
25	7		25	6		25	2		25	2
26	7		26	4		26	1		26	4
27	9		27	4		27	5		27	1
28	4		28	3		28	4		28	2
29	6		29	3		29	2		29	2
30	7		30	8		30	3		30	1
31	9		31	7		31	5		31	1
32	5		32	6		32	3		32	2
33	6		33	4		33	3		33	3
34	6		34	4		34	7		34	0
35	6		35	3		35	6		35	0
36	4		36	4		36	3		36	2
37	6		37	5		37	3		37	4
38	3		38	7		38	5		38	0
39	3		39	4		39	4		39	1
40	9		40	5		40	0		40	1
41	3		41	1		41	1		41	3
42	6		42	5		42	4		42	0
43	2		43	3		43	0		43	1

44	5		44	5		44	1		44	2
45	6		45	4		45	0		45	0
46	3		46	4		46	3		46	0
47	5		47	4		47	3		47	2
48	4		48	5		48	6		48	1
49	7		49	6		49	3		49	1
50	5		50	2		50	4		50	0
51	9		51	7		51	5		51	1
52	5		52	6		52	3		52	2
53	6		53	4		53	3		53	3
54	6		54	4		54	7		54	0
55	6		55	3		55	6		55	0
56	4		56	4		56	3		56	2
57	6		57	5		57	3		57	4
58	3		58	7		58	5		58	0
59	3		59	4		59	4		59	1
60	9		60	5		60	0		60	1
61	3		61	1		61	1		61	3
62	6		62	5		62	4		62	0
63	2		63	3		63	0		63	1
64	5		64	5		64	1		64	2
65	6		65	4		65	0		65	0
66	3		66	4		66	3		66	0
67	5		67	4		67	3		67	2
68	4		68	5		68	6		68	1
69	7		69	6		69	3		69	1
70	5		70	2		70	4		70	0
71	6		71	6		71	1		71	3
72	4		72	5		72	5		72	1
73	7		73	4		73	3		73	3
74	6		74	6		74	2		74	2
75	3		75	8		75	4		75	3
76	4		76	7		76	5		76	2
77	5		77	6		77	3		77	3
78	4		78	7		78	2		78	1
79	6		79	5		79	1		79	1
80	5		80	6		80	2		80	3

Descriptive statistics

Groups	N	Mean	Std. Deviation	Std. Error of Mean
CG1	80	5.3875	2.00943	.22466
CG2	80	4.8500	1.55164	.17348
EG1	80	3.2000	1.81694	.20314

EG2	80	1.7125	1.22416	.13687
Total	320	3.7875	2.20824	.12344

ANOVA

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	667.175	3	222.392	79.106	.000
Within Groups	888.375	316	2.811		
Total	1555.550	319			

Multiple Comparisons

(I) Groups	(J) Groups	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval	
					Lower Bound	Upper Bound
CG1	CG2	.53750*	.26511	.043	.0159	1.0591
	EG1	2.18750*	.26511	.000	1.6659	2.7091
	EG2	3.67500*	.26511	.000	3.1534	4.1966
CG2	CG1	-.53750*	.26511	.043	-1.0591	-.0159
	EG1	1.65000*	.26511	.000	1.1284	2.1716
	EG2	3.13750*	.26511	.000	2.6159	3.6591
EG1	CG1	-2.18750*	.26511	.000	-2.7091	-1.6659
	CG2	-1.65000*	.26511	.000	-2.1716	-1.1284
	EG2	1.48750*	.26511	.000	.9659	2.0091
EG2	CG1	-3.67500*	.26511	.000	-4.1966	-3.1534
	CG2	-3.13750*	.26511	.000	-3.6591	-2.6159
	EG1	-1.48750*	.26511	.000	-2.0091	-.9659

*. The mean difference is significant at the 0.05 level.